Assignment 1

October 24, 2024

Problem 1

Consider the differential equation $dy/dx = (y-3)^2$.

1.

Find the general solution of the differential equation and the singular solution that is not included in the general solution.

Feedback:

du behøver ikke vedhæfte kode

plot også den singulære løsning og angiv asymptoten med en anden farve

First the general solution is found by solving it as a seperable differential equation:

$$
\begin{aligned}
\frac{dy}{dx} &= (y-3)^2 \\
\frac{1}{(y-3)^2} dy &= 1dx && y \neq 3 \\
\int \frac{1}{(y-3)^2}\, dy &= \int 1\, dx \\
-\frac{1}{y-3} &= x + C \\
y - 3 &= -\frac{1}{x+C} \\
y &= 3 - \frac{1}{x+C} && \text{(the general solution)}
\end{aligned}
$$

It can now be seen that since the right hand side of the general solution features 3 subtracted by a fraction. Since only denominator of the fraction contains variables and the numerator is not 0, the fraction will never be 0 and the solution will never reach

$$y = 3 - 0$$

This means that the singular solution that is not included in the general solution, which was found, is $y = 3$.

2.

Find and plot the solution for the initial value $y(0) = 4$. Comment on the solution.

Feedback:

du behøver ikke vedhæfte kode

plot også den singulære løsning og angiv asymptoten med en anden farve

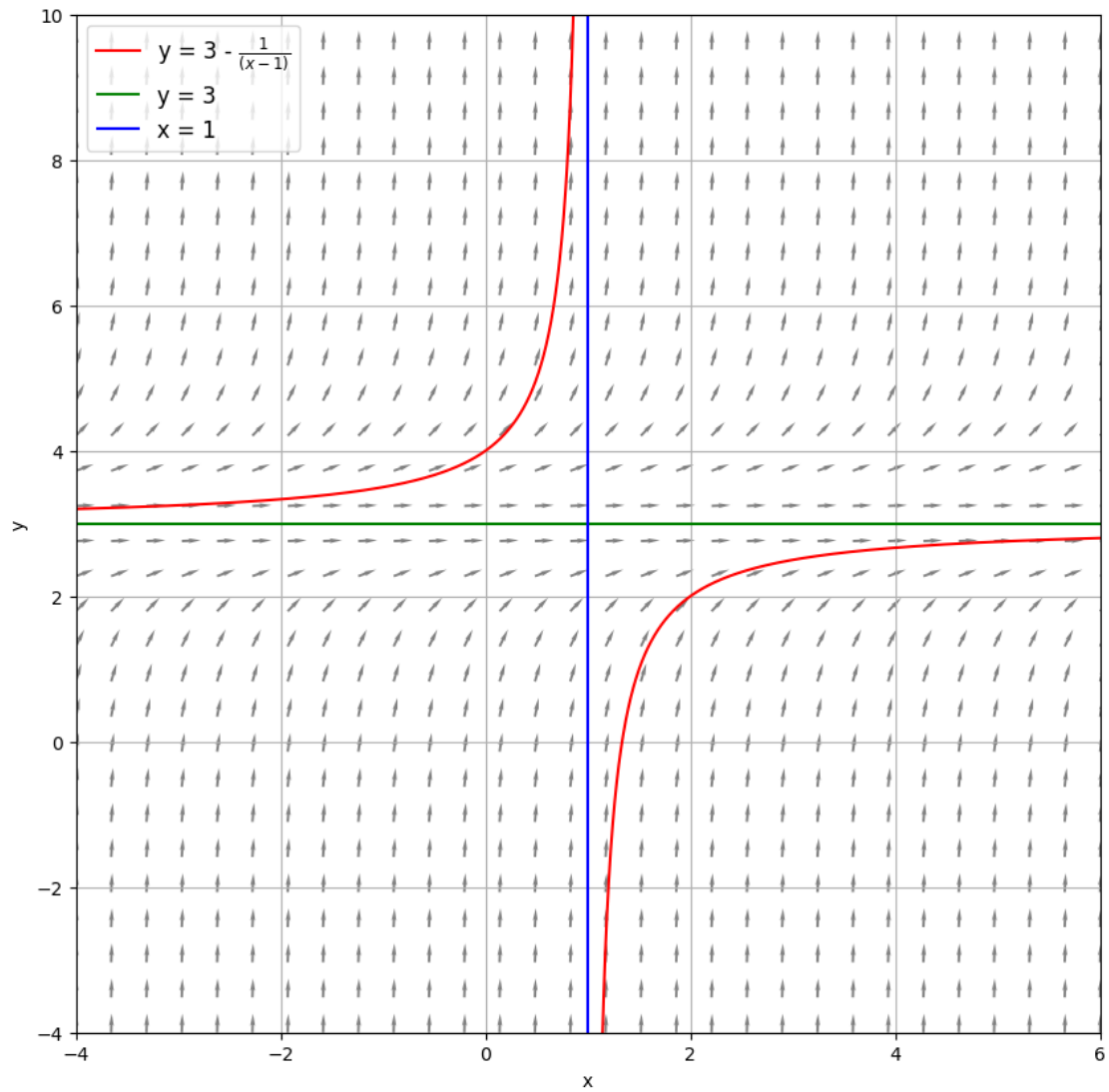
Since $y(0) \neq 3$, the inital value problem can be solved using the general solution from before. This is done by substituting y for 4 and x for 0.

$$
\begin{aligned}
4 &= 3 - \frac{1}{0 + C} \\
1 &= -\frac{1}{C} \\
C &= -1
\end{aligned}
$$

The solution for the intial value is then

$$y = 3 - \frac{1}{x - 1}$$

[30]:
```
<matplotlib.legend.Legend at 0x7436d6aa78b0>
```



The solution consists of two parts, which are disconnected by an asymptote at $x = 1$. This disconnection is due to the fact that the denominator in fraction of the solution would become 0 at $x = 1$. The solution function is therefore not defined there. When x approaches $-\infty$ the solution approaches $y = 3$, but it will never be reached since it is not defined for the general solution that was used. Toward $+\infty$ the function of the solution also approaches $y = 3$, but since the inital value problem was for $y(0)$, the solution only refers to the left side of asymptote and not the right side, which is a solution for a different initial value problem.

3.

What does Theorem 1, Existence and Uniqueness of Solutions say about the solution(s) of the initial value problem $y' = (y-3)^2, y(a) = b$ for arbitrary choices of a and b?

Feedback:

her mangler du at kommenterer på dy/dx = f(x,y) samt deres partielt afledte mht. y. Det er deres kontinuitet der dikterer om du er i stand til at finde en lokal unik løsning

The function defining the differential equation, $(y-3)^2$, is continuous on the rectangle defined on the open interval from negative infinity to positive infinity. Its partial derivative with respect to y is given as

$$\frac{d}{dy}(y-3)^2 = \frac{d}{dy}y^2 + 9 - 6y = 2y - 6$$

This function is also continuous on the same rectangle. This means that the given inital value problem satisfies the assumptions of Theorem 1, meaning that the following can be said about the solutions of the inital value problem.

The theorem states that for some open interval I there is one and only one solution for an initial value problem. Since we have found a solution that is not defined for x=1, the solution is defined only on the open interval $I = (-\infty, 1)$. This means that the right side of the function is not part of the solution, but that it is a solution to a different initial value problem, which is defined on the open interval $I = (1, \infty)$. This can be generalized by seeing that for any choice of a and b, the constant C can be described as

$$\begin{aligned} b &= 3 - \frac{1}{a+C} \\ b - 3 &= -\frac{1}{a+C} \\ (a+C)(b-3) &= -1 \\ a + C &= -\frac{1}{b-3} \\ C &= \frac{1}{3-b} - a \end{aligned}$$

The denominator of the solution would then approach 0 as x approaches $\frac{1}{b-3} + a$, which means that there will be an asymptote there. Like before the same function would then be two different solution on the open intervals $I = (-\infty, \frac{1}{b-3} + a)$ and $I = (\frac{1}{b-3} + a, \infty)$ respectively.

There is still the special case for $y(a) = 3$, which is not defined for the general solution. Here there is only a single function, $y(x) = 3$, counting as a single solution to the problem defined for all values of x in the open interval $I = (-\infty, \infty)$.

Problem 2

Consider the differential equation $y' = (y-2)\cos(x)$.

1.

Solve the differential equation by hand for the initial value $y(0) = 0$. Account for all intermediate steps in the solution.

Feedback:

for y forskellig fra 2

manger numerisk tegn (til ln)

(+-)eˆC1

First the general solution is found by solving it as a seperable differential equation:

$$\begin{aligned} \frac{dy}{dx} &= (y-2)\cos(x) \\ \frac{1}{y-2}dy &= \cos(x)dx && y \neq 2 \\ \int \frac{1}{y-2}\,dy &= \int \cos(x)\,dx \end{aligned}$$

Here substitution can be used to solve the integral on the left side, but it can also simply be solved by knowing the derivative $\frac{d}{dx}\ln(f(x)) = \frac{1}{f(x)} \cdot f'(x)$. Since the $\frac{d}{dy}y - 2 = 1$, the integral of $y-2$ simply becomes $\ln(y-2)$.

$$\begin{aligned} \ln(|y-2|) + C_y &= \sin(x) + C_x \\ \ln(|y-2|) &= \sin(x) + C_1 && C_1 = C_x - C_y \\ |y-2| &= e^{\sin(x)+C_1} \\ y-2 &= \pm e^{\sin(x)} \cdot e^{C_1} \\ y-2 &= C_2 e^{\sin(x)} && C_2 = \pm e^{C_1} \\ y &= C_2 e^{\sin(x)} + 2 && \text{(the general solution)} \end{aligned}$$

The initial value can then be inserted into the general solution to find C_2

$$\begin{aligned} 0 &= C_2 e^{\sin(0)} + 2 \\ -2 &= C_2 e^0 \\ C_2 &= -2 \end{aligned}$$

Inserting the value found for C_2 back into the general solution gives the particular solution for the initial value problem

$$y = -2e^{\sin(x)} + 2 \qquad \text{(the particular solution)}$$

2.

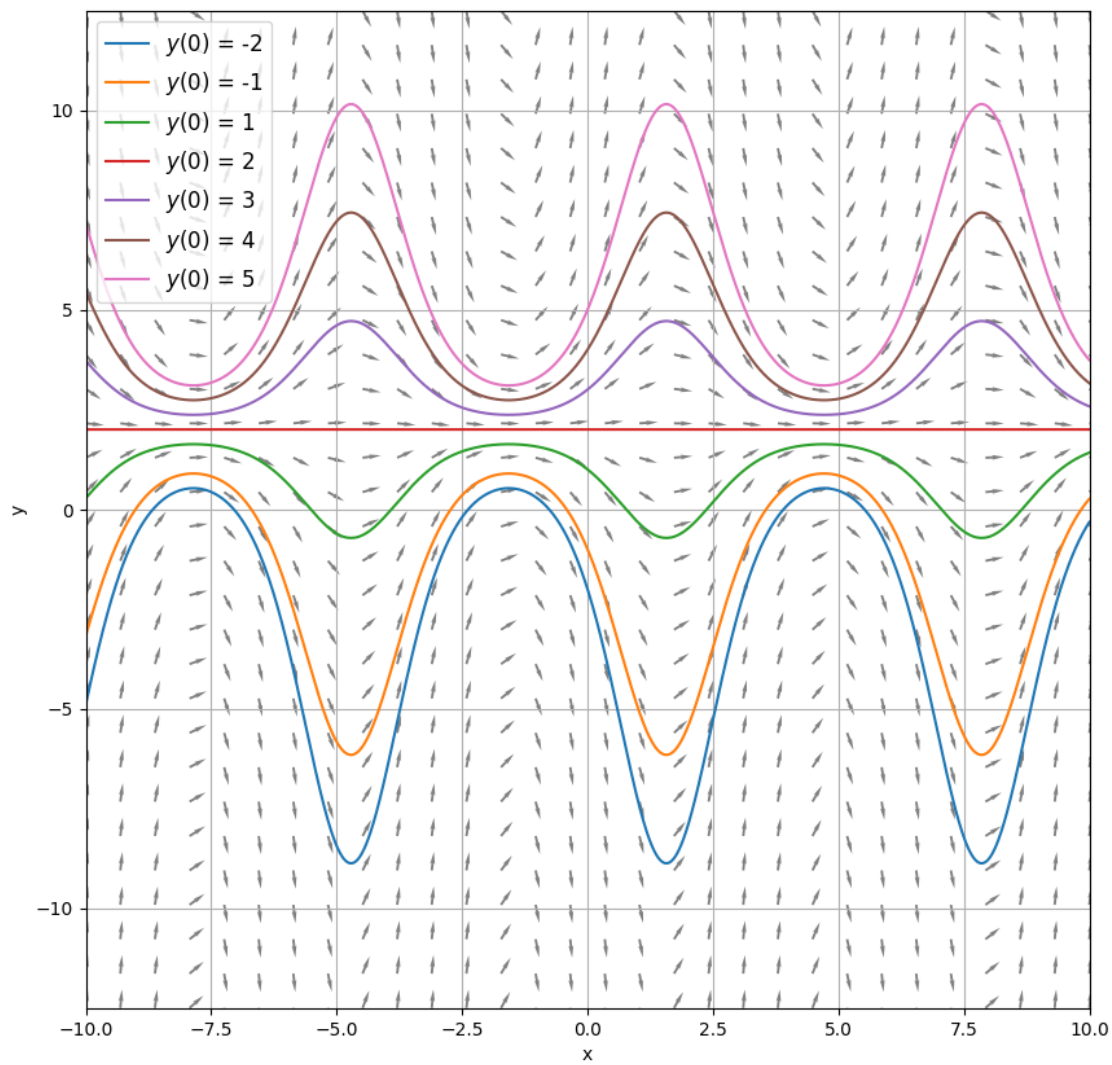
Discuss the solutions to the differential equation if other initial values than 0 are used, i.e., the $y(0) = b, b \neq 0$ case. Include plots of the slope field and representative solutions.

As can be seen when finding C_2 in the previous part, it only depends on the value of b since $e^{\sin(0)} = 1$. For any value b the constant C_2 can be found as

$$C_2 = b - 2$$

making it very easy to find solutions using for loops in python.

This also means that the solution for the initial value $y(0) = 2$, would become the constant function $y = (2-2)e^{\sin(x)} + 2 = 2$. Every other function however will grow and shrink exponentially with a periodicity defined by the sinus function in the exponent. Various solutions (excluding $y(0) = 0$) can be seen below.



Problem 3

Consider the differential equation $y' = 2x - y$.

1.

Solve the differential equation with the initial value $y(0) = 3$.

First the general solution is found. This is done by getting the equation on the form $dy/dx + P(x)y = Q(x)$:

$$\begin{aligned} \frac{dy}{dx} &= 2x - y \\ \frac{dy}{dx} + y &= 2x \end{aligned}$$

The general solution can now be found using Theorem 1.5.1:

$$\begin{aligned} y(x) &= e^{-\int 1\,dx}\left[\int \left(2xe^{\int 1\,dx}\right)\,dx + C\right] \\ &= e^{-x}\left[\int 2xe^{x}\,dx + C\right] \\ &= e^{-x}\left[2x \cdot e^{x} - \int 2e^{x}\,dx + C\right] && \text{(integration by parts)} \\ &= e^{-x}[2x \cdot e^{x} - 2e^{x} + C] \\ &= 2x - 2 + Ce^{-x} \\ &= Ce^{-x} + 2x - 2 && \text{(the general solution)} \end{aligned}$$

The constant C corresponding to the problem can be found by inserting the inital value:

$$\begin{aligned} 3 &= Ce^{-0} + 2 \cdot 0 - 2 \\ 3 &= C - 2 \\ C &= 5 \end{aligned}$$

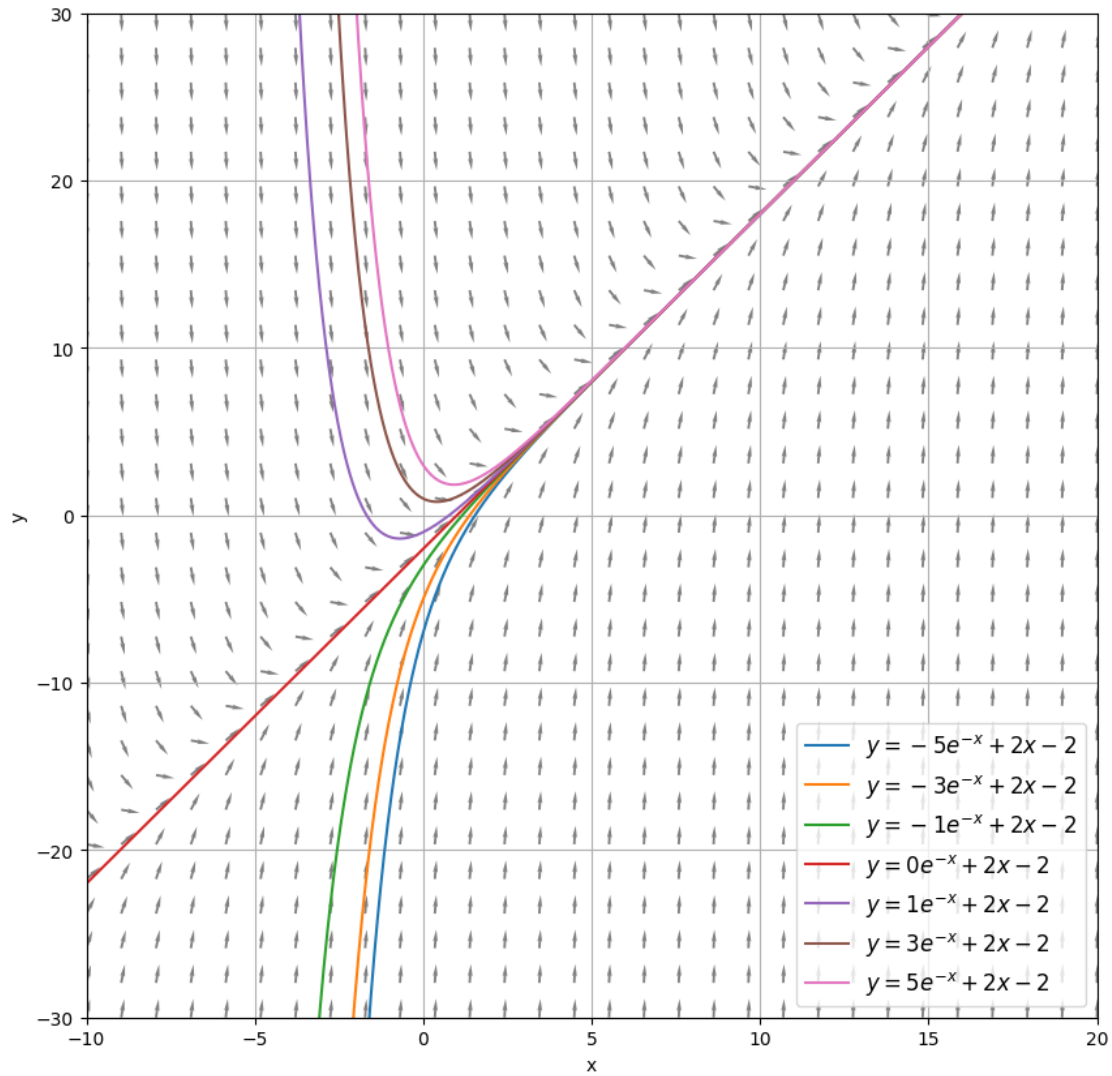
Inserting the value found for C back into the general solution gives the particular solution for the initial value problem

$$y = 5e^{-x} + 2x - 2 \qquad \text{(the particular solution)}$$

2.

Account for the asymptotic behaviour of the solution for this initial value as well as other initial values. Include any necessary plots.

Solutions with integer values of C between -5 and 5 are plotted below. The even integers aside from 0 are excluded to avoid cluttering the graph.



As can be seen in the graph above all the solutions have a vertical asymptote, which is shifted further to the right along the x-axis the larger the value of C is numerically. This makes sense, since Ce^{-x} will approach ∞ as x approaches $-\infty$. Something almost all solutions have in common is a asymptote defined by the function $y(x) = 2x - 2$. They all approach this function since the Ce^{-x} term approaches 0 as x approaches ∞, making the entire expression of any solution approach $y = 2x - 2$. Since the solution for $C = 0$ is $y(x) = 2x - 2$, this solution in particular does not have the asymptote, since it itself is the other solutions asymptote.

3.

Assume that the differential equation has been solved for two different initial values, $y_1(a) = b_1$ and $y_2(a) = b_2$ with $b_1 \neq b_2$.

Can the two solutions, $y_1(x)$ and $y_2(x)$, cross each other?

Based on the general solution we can let $y_1(x) = C_1 e^{-x} + 2x - 2$ and $y_2(x) = C_2 e^{-x} + 2x - 2$, where $C_1 \neq C_2$ since they are solutions to different initial values. We can set them to be equal to each other to see if they cross and if so for which value of x.

$$\begin{aligned} C_1 e^{-x} + 2x - 2 &= C_2 e^{-x} + 2x - 2 \\ C_1 &= C_2 \end{aligned}$$

Every term other than the constants cancel out leaving the expression as a contradiction, since they cannot be equal if they are solutions to different initial values. Thus, the two solutions will never cross.

Problem 4

Consider the logistic differential equation $y' = 2y(5 - y)$.

Feedback:

For hvilke y gælder dette?

mangler numrisk tegn (til ln)

(+-)eˆ10C1

Plottet kunne godt være begrænset lidt mere i x, så den logistiske vækst kom lidt mere til syne

1.

Compute and plot the solutions for $y(0) = 1$ and $y(1) = 1$. What features do the two solutions have in common?

The general solution is first calculated as a seperable differential equation:

$$\begin{aligned} \frac{dy}{dx} &= 2y(5-y) \\ \int \frac{1}{2y(5-y)}\, dy &= \int 1\, dx \qquad\qquad y \neq 0 \wedge y \neq 5 \\ \frac{1}{2} \int \frac{1}{y(5-y)}\, dy &= x + C_1 \end{aligned}$$

The left side can be rewritten using partial fractions as follows

$$
\begin{aligned}
\frac{A}{y} + \frac{B}{5-y} &= \frac{1}{y(5-y)} \\
\frac{A \cdot y(5-y)}{y} + \frac{B \cdot y(5-y)}{5-y} &= 1 \\
A \cdot (5-y) + B \cdot y &= 1 \\
A \cdot 5 + B \cdot 0 &= 1 \qquad y = 0 \text{ is chosen to find A since any y value is valid} \\
A &= \frac{1}{5} \\
A \cdot 0 + B \cdot 5 &= 1 \qquad y = 5 \text{ is chosen to find B} \\
B &= \frac{1}{5}
\end{aligned}
$$

Inserting the partial fractions back in the caluculation yields

$$
\begin{aligned}
\frac{1}{2} \int \frac{1}{5y} + \frac{1}{5(5-y)} \, dy &= x + C_1 \\
\frac{1}{10} \int \frac{1}{y} + \frac{1}{5-y} \, dy &= x + C_1 \\
\frac{1}{10} \cdot (\ln(|y|) - \ln(|5-y|)) &= x + C_1 \\
\ln(|y|) - \ln(|5-y|) &= 10x + 10C_1 \\
e^{\ln(|y|) - \ln(|5-y|)} &= e^{10x + 10C_1} \\
\left| \frac{y}{5-y} \right| &= e^{10x} \cdot e^{10C_1} \\
\frac{y}{5-y} &= \pm e^{10x} \cdot e^{10C_1} \\
y &= e^{10x} \cdot C_2 \cdot (5-y) \qquad C_2 = \pm e^{10C_1} \\
y &= e^{10x} \cdot C_2 \cdot 5 - e^{10x} \cdot C_2 \cdot y \\
y + e^{10x} \cdot C_2 \cdot y &= 5 \cdot e^{10x} \cdot C_2 \\
y \cdot (1 + e^{10x} \cdot C_2) &= 5 \cdot e^{10x} \cdot C_2 \\
y &= \frac{5 \cdot e^{10x} \cdot C_2}{1 + e^{10x} \cdot C_2} \\
y &= \frac{5 \cdot e^{10x} \cdot \frac{1}{C_3}}{1 + e^{10x} \cdot \frac{1}{C_3}} \qquad C_3 = \frac{1}{C_2} \\
y &= \frac{5 \cdot e^{10x}}{C_3 + e^{10x}} \qquad \text{(the general solution)}
\end{aligned}
$$

The initial value $y(0) = 1$ can now be inserted into the general solution:

$$
\begin{aligned}
1 &= \frac{5 \cdot e^{10 \cdot 0}}{C_3 + e^{10 \cdot 0}} \\
C_3 + 1 &= 5 \\
C_3 &= 4
\end{aligned}
$$

This means that the solution for the initial value $y(0) = 1$ is $y_0 = \frac{5 \cdot e^{10x}}{4 + e^{10x}}$.

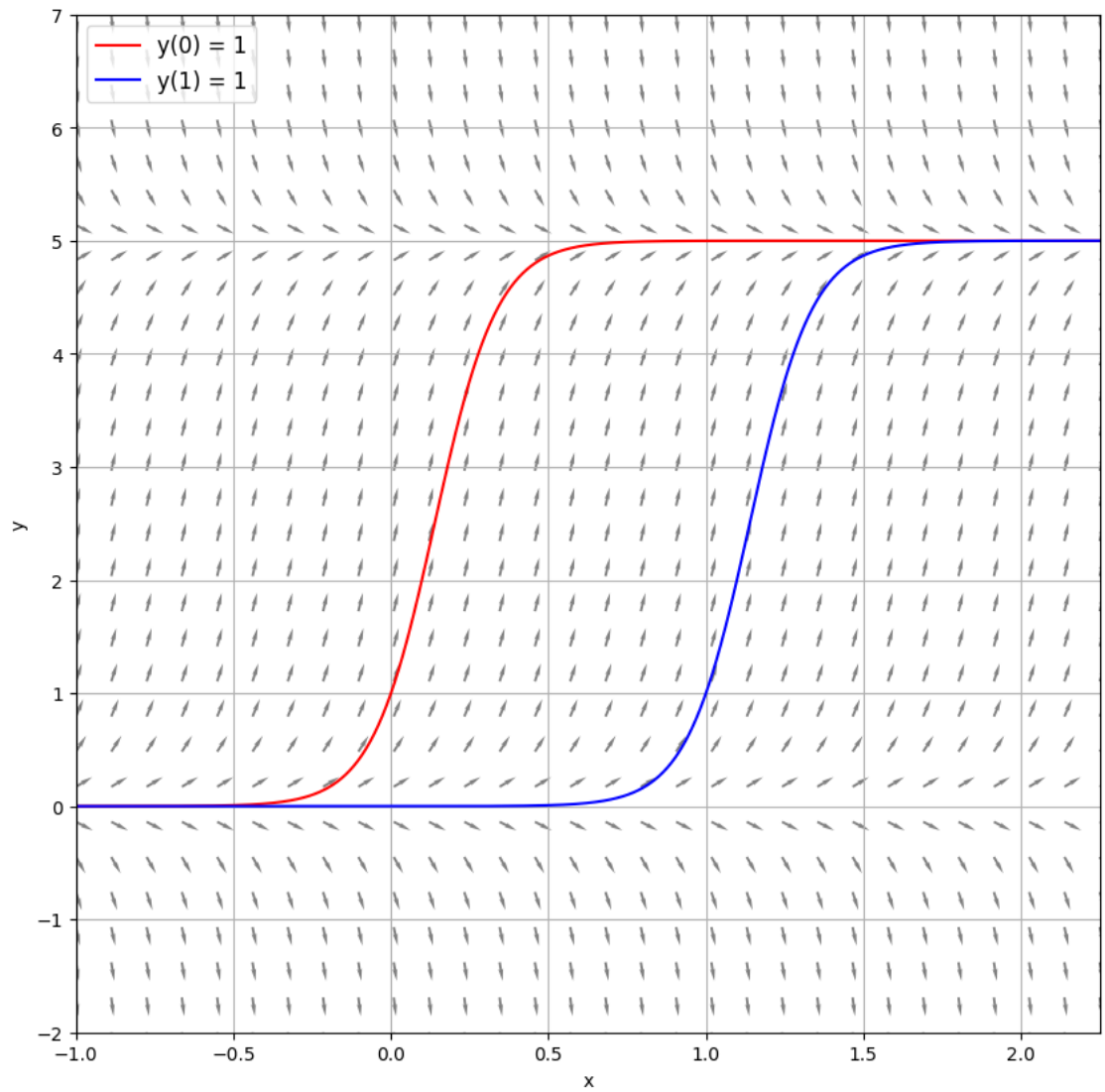
Now inserting the initial value $y(1) = 1$ yields:

$$
\begin{aligned}
1 &= \frac{5 \cdot e^{10 \cdot 1}}{C_3 + e^{10 \cdot 1}} \\
C_3 + e^{10} &= 5 \cdot e^{10} \\
C_3 &= 5 \cdot e^{10} - e^{10} \\
C_3 &= 4 \cdot e^{10}
\end{aligned}
$$

Giving the solution $y_1 = \frac{5 \cdot e^{10x}}{4 \cdot e^{10} + e^{10x}}$

These two solutions can now be plotted in python

[33]:
```
<matplotlib.legend.Legend at 0x7436d66d0400>
```



Both functions have the same sigmoid shape, but that the solution for $y(1) = 1$ is shifted to the right. It makes sense that the functions approach 0 toward $-\infty$ since e^{10x} approaches 0 and the entire expression would approach $\frac{0}{C}$. When approaching $+\infty$, the constant in the denominator becomes obsolete, meaning the expression approaches $\frac{5e^{10x}}{e^{10x}}$. The two e^{10x} cancels out leaving 5, which is also seen in the graph. The reason for the shift is that the constant in the denominator does make a difference at small x-values, meaning that it takes a longer time before the e^{10x} terms explode the bigger the constant is.

2.

Assuming $y(0) = 1$, when is $y(x) = 2$?

Feedback:

skriv den approksimerede værdi også

Since inital value $y(0) = 1$ has already been solved for, the particular solution is given as $y(x) = \frac{5 \cdot e^{10x}}{4+e^{10x}}$. A simple equation can now be made and solved by setting the solution equal to 2 and solving for x.

$$\begin{aligned}
2 &= \frac{5 \cdot e^{10x}}{4 + e^{10x}} \\
8 + 2e^{10x} &= 5e^{10x} \\
8 &= 5e^{10x} - 2e^{10x} \\
8 &= e^{10x} \cdot (5 - 2) \\
e^{10x} &= \frac{8}{3} \\
10x &= \ln\left(\frac{8}{3}\right) \\
x &= \frac{\ln\left(\frac{8}{3}\right)}{10}
\end{aligned}$$

Assuming $y(0) = 1$, the solution will reach 2 at $x = \frac{\ln\left(\frac{8}{3}\right)}{10} \approx 0.098$.

3.

Assume that we use the logistic differential equation to model a population and that an additional term is added to the equation to model harvesting, i.e, $y' = 2y(5 - y) - h, h > 0$.

What is the maximum value of h for which the population still exists?

For the population to still exist, the harvest must either make the population grow (positive slope) or stay at a constant (slope of 0). It can be seen that the growth of the population is modelled after a quadratic equation depending on the current population (y) and the harvesting (h)

$$\frac{dy}{dx} = 2y(5 - y) = -2y^2 + 10y - h$$

Since the first term ($-2y^2$) is negative, it must mean that the vertex of the model is also the maximum point. The constant h will move the vertex toward $-\infty$ on the second axis as h approaches ∞. When the vertex gets below $\frac{dy}{dx} = 0$, it means that the population can no longer grow nor stay at a constant, meaning that it will cease to exist. The maximal value of h is then the value that places the vertex at exactly $\frac{dy}{dx} = 0$. In other words the quadratic equation must only have a single root, which is true if the discriminant of the equation is 0. The discriminant can then be set to 0 and the resulting equation can be solved for h.

$$\begin{aligned}
0 &= b^2 - 4ac \\
0 &= 10^2 - 4 \cdot (-2) \cdot (-h) \\
0 &= 100 - 8h \\
h &= \frac{100}{8} = 12.5
\end{aligned}$$

For the population to still exist, the harvesting constant can have a maximal value of $h = 12.5$.

Problem 5

Consider a system modelled by the differential equation

$$\frac{dy}{dx} = M(x) - 2y,\ M(x) = \begin{cases} 5, & 0 \leq x \leq 5 \\ 0, & x > 5 \end{cases}$$

1.

Solve the differential equation for the initial value $y(0) = 1$ and plot the solution for $0 < x < 10$.

First the general solution is found. This is done by getting the equation on the form $dy/dx + P(x)y = Q(x)$:

$$\begin{aligned} \frac{dy}{dx} &= M(x) - 2y \\ \frac{dy}{dx} + 2y &= M(x) \end{aligned}$$

The general solution can now be found using Theorem 1.5.1. This is first done for the interval $0 \leq x \leq 5$:

$$\begin{aligned} y_1(x) &= e^{-\int 2\,dx} \left[\int \left(5e^{\int 2\,dx} \right) dx + C_1 \right] \\ &= e^{-2x} \left[5 \int e^{2x}\,dx + C_1 \right] \\ &= e^{-2x} \left[5 \cdot \frac{1}{2} \cdot e^{2x} + C_1 \right] \\ &= \frac{5}{2} + C_1 e^{-2x} && \text{(general solution for 0 x 5)} \end{aligned}$$

It is then found for for $5 < x < 10$

$$\begin{aligned} y_2(x) &= e^{-\int 2\,dx} \left[\int \left(0e^{\int 2\,dx} \right) dx + C_2 \right] \\ &= e^{-2x} \left[C_2 \right] \\ &= C_2 e^{-2x} && \text{(general solution for 5 < x < 10)} \end{aligned}$$

The initial value is then inserted into the first general solution:

$$\begin{aligned} 1 &= \frac{5}{2} + C_1 e^{-2 \cdot 0} \\ C_1 &= -\frac{3}{2} \end{aligned}$$

The particular solution for the interval $0 \leq x \leq 5$ is therefore:

$$y(x) = \frac{5}{2} - \frac{3}{2}e^{-2 \cdot x}$$

To ensure the continuity between the part before and after $x = 5$, the particular solutioon $y_1(x)$ can be set equal to $y_2(x)$ at $x = 5$. The particular solution for the interval $5 \leq x \leq 10$ is then found by isolating C_2:

$$\begin{aligned} C_2 e^{-2 \cdot 5} &= \frac{5}{2} - \frac{3}{2}e^{-2 \cdot 5} \\ C_2 &= \frac{5}{2}e^{10} - \frac{3}{2} \end{aligned}$$

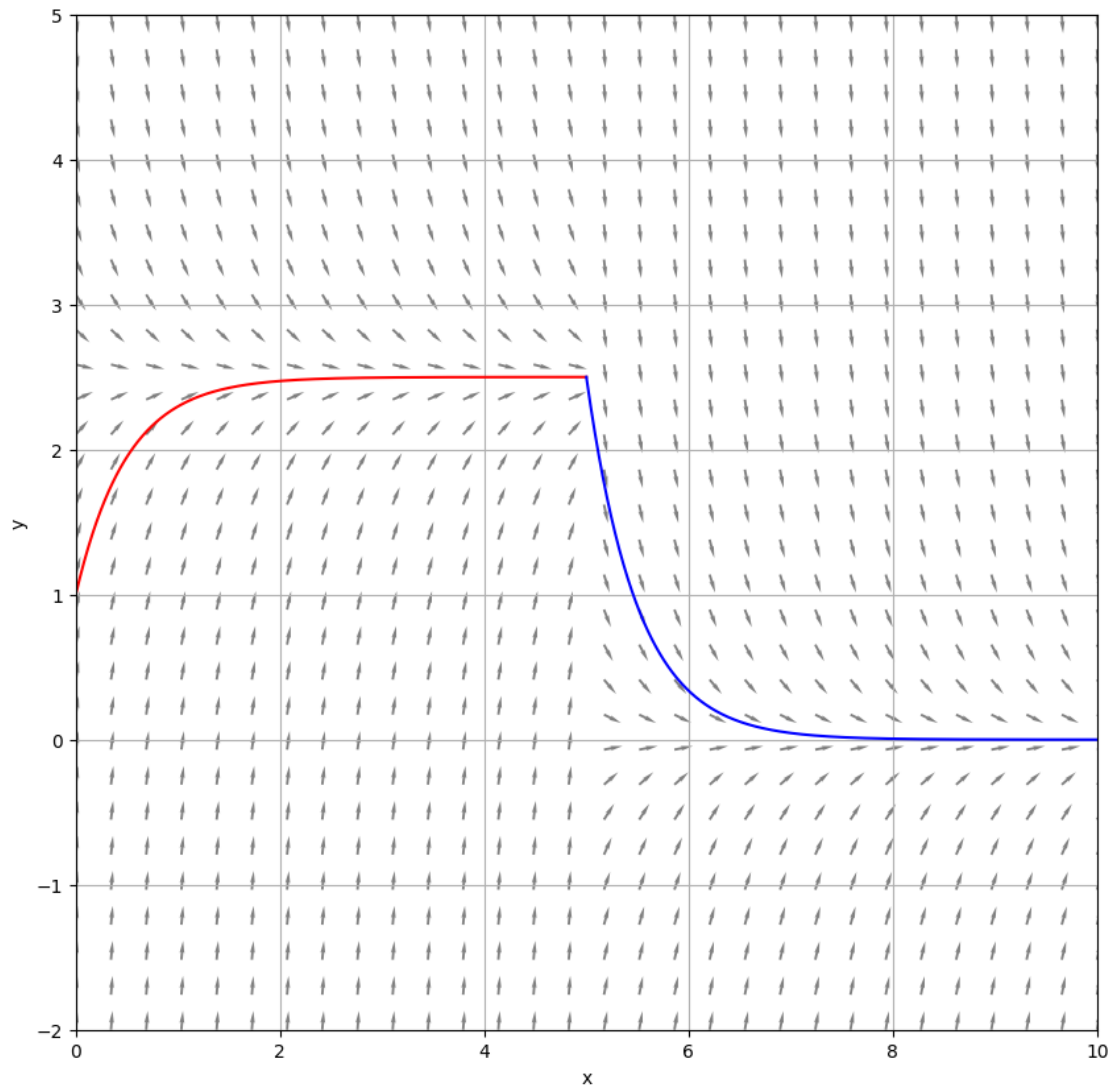
The new constant is then inserted into $y_2(x)$:

$$\begin{aligned} y_2(x) &= (\frac{5}{2}e^{10} - \frac{3}{2})e^{-2x} \\ &= \frac{5}{2}e^{10-2x} - \frac{3}{2}e^{-2x} \end{aligned}$$

The combined particular solution would then be:

$$y(x) = \begin{cases} \frac{5}{2} - \frac{3}{2}e^{-2 \cdot x}, & 0 \leq x \leq 5 \\ \frac{5}{2}e^{10-2x} - \frac{3}{2}e^{-2x}, & x > 5 \end{cases}$$

[34]:
```
[<matplotlib.lines.Line2D at 0x7436d6480b20>]
```



2.

In the above, $M(x)$ is given as a piece-wise defined function. Give a real world example of modeling a system where the inclusion of a piece-wise defined function makes sense.

A piece-wise function would a good model if a system had some kind of switch, which would change the behavior of the system. In the case of the model given in the exercise, it would match the behavior of a capacitor that is charged and discharged. The red part of the graph would be the capacitor charging at an exponential rate, and the blue part would be the capacitor exponentially dischargning. In this model, the y-axis, which would be voltage, starts at 1 and not 0 which is not typical, but it could just be due to a weird choice of frame.